



# Rishabh Thakur

Roll No: 250101084

B.Tech in Computer Science and Engineering

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## EDUCATION

| Degree/Certificate | Institute/Board                          | CGPA/Percentage | Year         |
|--------------------|------------------------------------------|-----------------|--------------|
| B.Tech.            | Indian Institute of Technology, Guwahati | 7.75 (Current)  | 2025-Present |
| Senior Secondary   | CBSE Board                               | 96.6%           | 2025         |
| Secondary          | CBSE Board                               | 97.8%           | 2023         |

## EXPERIENCE

- **Stiri Private Limited** Jan 2026 – Present  
*Co-Founder & Lead Developer* Guwahati, India
  - Architected and built the **complete full-stack platform** from scratch using **React 19**, **TypeScript**, **Express.js**, and **Supabase** (**PostgreSQL RLS** & **pgvector** for vector search).
  - Engineered client-side edge AI using **ONNX Runtime (WASM)** for YOLOv8 garment detection and integrated **Google Gemini API**; managed startup operations, product roadmap, Razorpay onboarding, and cloud deployment.
- **Internet Archive (Open Library)** Jul. 2026 – Present  
*Open Source Contributor* Remote / Open Source
  - Engineered **FastAPI backend endpoints** for profile picture management and patron check-in validation; resolved critical security dependency crashes on core authentication paths.
  - Designed smart edition-ranking algorithms in **Python** (`get_best_edition`) and fixed regex parsing bugs in Volumes Multiget JSON APIs.

## PROJECTS

- **Velora – Corporate Mobility Optimization Platform** Jan. 2026 – Feb. 2026  
*Group Project, Kriti 2026, Inter Hostel Technical Competition, IITG*  GitHub
  - Developed a full-stack corporate mobility platform using **Django**, **React** + **Vite**, and **Capacitor Android** with C++ executable orchestration and **OSRM route geometry**.
- **Momentum Trading Strategy – Quantitative Finance** Jan. 2026 – Feb. 2026  
*Kriti 2026, Inter Hostel Technical Competition, IITG*  GitHub
  - Designed a systematic long-only equity momentum strategy for NSE 500 using multi-horizon momentum (1m–12m) achieving **17.70% CAGR** vs benchmark **8.62% (Sharpe 0.92)** using **Numba JIT** (~10× speedup).
- **IRCTC Booking Automation** Feb. 2026  
*Personal Project*  GitHub
  - Engineered high-reliability booking automation using **Playwright** with login time-gating, multi-strategy captcha solving (**EasyOCR**, **Google Vision API**), and UPI payment handoff (<30s execution).

## ACHIEVEMENTS

- **Codeforces**: Maximum rating – **1503** | **Specialist** | Handle: *RsbhThakur1*. 2026
- **Kriti 2026 Gamedev**: Secured **Rank 2** out of all hostels in Inter-Hostel Tech Competition . 2026
- **JEE**: Secured **AIR 652** in JEE Advanced and **AIR 852** in JEE Mains among 1.4M+ candidates. 2025
- **CMI & ISI**: In **Top 40** merit list for CMI Entrance and **Top 180** national merit list for ISI Entrance. 2025
- **National Olympiads**: In **Top 115** at ZIO and **Top 350** at NSEP nationally (qualified for **INOI** and **INPhO**). 2025
- **RMO & INMO**: Ranked **Delhi Top 6** (RMO 2024) with **80 marks**; qualified for **INMO 2024 & 2025**. 2023–2025

## TECHNICAL SKILLS

- **Languages**: C, C++, Python, JavaScript, TypeScript, Java, SQL, HTML/CSS
- **Databases & Infrastructure**: PostgreSQL, Upstash Redis, MongoDB, Docker, Linux, Pytest, GitHub Actions, Git
- **Web & Frameworks**: React, Node.js, Express.js, Django, FastAPI, Pydantic, Playwright, Tailwind CSS
- **AI, ML & Edge Computing**: ONNX Runtime (WASM), MediaPipe, pgvector, NumPy, Pandas, Numba,  $\text{\LaTeX}$

## KEY COURSES TAKEN

- **Computer Science**: Data Structures & Algorithms, Computer Systems and Organization, Digital Design, Object Oriented Programming, Design and Analysis of Algorithms, Optimization, Formal Languages and Automata Theory
- **Mathematics**: Number Theory & Algebra, Probability Theory & Random Processes, Discrete Mathematics, Linear Algebra, Real Analysis

## POSITIONS OF RESPONSIBILITY

- **Deputy Coordinator**, Coding Club, IIT Guwahati Jan. 2026 – Present
  - Architected the **Problem of the Day (POTD)** engine and **Codeforces API Sync Engine** for **Coding Club Website (CCW)** using **Redis caching**, Agenda workers, and dense-ranking leaderboards.
  - Optimized **CourseHub** backend allotment caching for instant logins, engineered an **ImageKit CDN pipeline** for OneDrive media assets, and built the **Habit-IITG** room-cleaning backend for Hostel Affairs Board, IIT Guwahati.